

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHM

Out of Boundary Paths

1. Aim

To find the number of ways a ball can move outside a grid within a given number of moves using **Dynamic Programming**.

2. Algorithm

1. Start from the given position.
2. Move in four directions: up, down, left, and right.
3. If the ball goes outside the grid, count that path.
4. Otherwise, continue moving.
5. Repeat for the given number of moves.

3. Pseudocode

Start

Set answer = 0

For each possible move:

 Check up, down, left and right

 If position is outside:

 Increase answer

 Else:

 Store the number of ways

Return answer

Stop

4. Simple Python Code

```
def findPaths(m, n, moves, r, c):  
    dp = [[0] * n for _ in range(m)]  
    dp[r][c] = 1  
    answer = 0  
  
    for k in range(moves):  
        new = [[0] * n for _ in range(m)]  
        for i in range(m):  
            for j in range(n):  
                if dp[i][j] == 0:  
                    continue  
  
                # Up  
                if i == 0:  
                    answer += dp[i][j]  
                else:  
                    new[i-1][j] += dp[i][j]  
  
                # Down  
                if i == m-1:  
                    answer += dp[i][j]  
                else:  
                    new[i+1][j] += dp[i][j]  
  
                # Left  
                if j == 0:  
                    answer += dp[i][j]  
                else:  
                    new[i][j-1] += dp[i][j]  
  
                # Right  
                if j == n-1:  
                    answer += dp[i][j]
```

```

        else:
            new[i][j+1] += dp[i][j]

    dp = new

    return answer

print(findPaths(2, 2, 2, 0, 0))

```

5. Output

6

6. Output Screenshot

The screenshot shows an IDE window titled 'DAA.py - C:/Users/karth/AppData/Local/Programs/Python/Python314/DAA.py (3.14.7)'. The code defines a function `findPaths(m, n, moves, r, c)` that calculates the number of paths from a starting point (r, c) to the bottom-right corner (m-1, n-1) using a dynamic programming table `dp`. The function iterates over moves (Up, Down, Left, Right) and updates the `dp` table. The final result is printed as `print(findPaths(2, 2, 2, 0, 0))`.

An 'IDLE Shell 3.14.7' window is open, showing the execution output:


```

Python 3.14.7 (tags/v3.14.7:823f032, Aug 5 2026, 10:51:32) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/karth/AppData/Local/Programs/Python/Python314/DAA.py =====
>>> 6
>>>
    
```

7. Complexity Analysis

- **Time Complexity:** $O(m \times n \times \text{moves})$
- **Space Complexity:** $O(m \times n)$

8. Result

The number of paths going out of the boundary was successfully calculated using **Dynamic Programming**.